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MIL-L-53110(ME)
12 September 1991

MILITARY SPECIFICATION

LABORATORY, PETROLEUM, MODULAR BASE

This specification is approved for use within the USA Belvoir Research Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the modular base petroleum laboratory fully equipped with testing equipment, supplies and chemicals to perform quality surveillance activities on large quantities of petroleum-derived products, antirust and antifreeze additives in a theater of operations for 60 days.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

W-P-115	- Panel, Power Distribution.
W-R-550	- Rod, Ground (with Attachments).
TT-P-664	- Primer Coating, Synthetic, Rust-Inhibiting, Lacquer Resisting.
PPP-B-601	- Boxes, Wood, Cleated Plywood.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6640

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PPP-B-1055 - Barrier Material, Waterproof, Flexible.
PPP-C-1752 - Cushioning Material, Packaging,Unicellular Polyethylene Foam, Flexible.
PPP-T-60 - Tape: Packaging, Waterproof.

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-P-514 - Plate, Identification, Instruction and Marking, Blank.
MIL-T-704 - Treatment and Painting of Materiel.
MIL-B-7883 - Brazing of Steels, Copper, Copper Alloys, Nickel Alloys, Aluminum and Aluminum Alloys.
MIL-C-22992 - Connectors, Plugs, and Receptacles, Electrical, Waterproof, Quick Disconnect, Heavy Duty Type, General Specifications for.
MIL-C-46168 - Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-S-46756 - Semitrailer S Van, 10 Ton, 4-Wheel, Series.
MIL-L-51050 - Laboratory, Petroleum, Semitrailer Mounted.
MIL-C-53039 - Coating, Aliphatic Polyurethane, Single Component, Chemical Agent Resistant.
MIL-L-53111 - Laboratory, Petroleum, Supplemental, Semitrailer Mounted.
MIL-V-62038 - Vehicle, Wheeled, Penetration for Shipment and Storage.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standard for Federal Services.
FED-STD-313 - Material Safety Data Sheets, Preparation and Submission of.

MILITARY

MIL-STD-101 - Color Code for Pipelines and for Compressed Gas Cylinders.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U.S. Military Property.
MIL-STD-209 - Military Standard Slinging and Tiedown Provisions for Lifting and Tying Down of Military Equipment.
MIL-STD-454 - Standard General Requirements for Electronic Equipment.
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines.
MIL-STD-889 - Dissimilar Metals.
MIL-STD-1366 - Materiel Transportation System Dimensional and Weight Constraints, Definition of.

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MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1474	- Noise Limits for Army Materiel.
MIL-STD-1791	- Designing for Internal Aerial Delivery in Fixed Wing Aircraft.
MIL-STD-1857	- Grounding, Bonding and Shielding Practices.
MS39347	- 5940 Terminal, Post, Service & Ground.

(Unless otherwise indicated, copies of federal and military specifications and standards are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents and drawings. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

ME

TA13226E6500	- Semitrailer Mounted Petroleum Laboratory.

COMPONENTS PARTS LISTS

SC6640-97-CL-E02	- Mobile, Semitrailer Mounted Petroleum Laboratory Components Parts List.
90106	- Supplemental Semitrailer Mounted Petroleum Laboratory Components Parts List.

(Copies of drawings and components parts lists required by contractors in connection with specific acquisition functions should be obtained from the U.S. Army Belvoir Research, Development, and Engineering Center, ATTN: STRBE-FSP, Fort Belvoir, VA 22060-5606.)

DEPARTMENT OF LABOR (DOL)

29CFR1910	- Occupational Safety and Health Administration (OSHA).

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

DEPARTMENT OF TRANSPORTATION (DOT)

49CFR171-190	- DOT Hazardous Materials Regulations.
49CFR393	- DOT Federal Motor Carrier Safety Regulations (FMCSR).

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D.C. 20402.)

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2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN WELDING SOCIETY (AWS)

D1.1 - Structural Welding Code - Steel.
D1.2 - Structural Welding Code - Aluminum.

(Application for copies should be addressed to the American Welding Society, 550 N.W. LeJeune Road, P.O. Box 351040, Miami, FL 33135.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

501C - Fire Safety Criteria for Recreational Vehicles.
70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, 345 East 47th Street, New York, NY 10017.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

J429 - Mechanical and Material Requirements for Externally Threaded Fasteners.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

(Non-Government standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for associated detail specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

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3.1 Description. The modular base petroleum laboratory (hereinafter called "MBPL") shall be comprised of the semitrailer mounted petroleum laboratory (hereinafter called "unit one"), the supplemental semitrailer mounted petroleum laboratory (hereinafter called "unit two"), the power distribution module (PDM) and a 60 KW DED generator set, trailer mounted. Units one and two shall be compatible with an M818, 5 Ton, 6x6 truck tractor, (or equal). Unit one shall be in accordance with TA13226E6500 and MIL-L-51050. Unit two shall be in accordance with MIL-L-53111. Unit two shall be equipped with a catwalk (NSN 2510-01-205-5990) that will allow the two units to be coupled together by attaching the catwalk to the rear platform of both vehicles. The power distribution module shall be as defined herein.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Materials. The material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration prevention and control. Each unit of the MBPL shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operation and storage environments to which the MBPL may be exposed.

3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-T-704 and MIL-STD-889.

3.3.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish, or treatment used during fabrication of any component or subcomponent and shall make this information available, upon request, to the contracting officer or designated representative.

3.3.4 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces, and parts incorporated in the MBPL may be newly fabricated from recovered materials to the maximum extent practicable, provided the MBPL produced meets all other requirements of this specification. Used, rebuilt, or remanufactured components, pieces, and parts shall not be incorporated in the MBPL.

3.4 Design. Each unit of the MBPL shall be designed within the parameters and limitations herein. The design shall limit or prevent conditions hazardous to personnel or deleterious to equipment. Units one and two shall comply with the Occupational Safety and Health Administration (OSHA) standards in 29CFR1910 and the utility systems within each unit shall comply with the

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mobile vehicle standards in NFPA-501C and NFPA-70. No deviation from the prescribed limitations or equipment specifications is permissible without prior approval of the contracting officer. Any data (e.g., shop and fabrication drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a supplier to support fabrication and manufacture of the production item or unit, shall be made available, upon request, for inspection by the contracting officer or designated representative.

3.4.1 Semitrailer design modifications. The semitrailers for unit one and unit two of the MBPL shall be modified to the extent defined in TA13226E6500 for unit one and MIL-L-53111 for unit two. Floor, wall, and ceiling fabrication shall be in accordance with MIL-L-51050, for unit one, and MIL-L-53111, for unit two.

3.4.2 Cabinetry. Units one and two of the MBPL shall have 1-inch thick solid epoxy (or equivalent) countertops, with metal cabinetry below, on the roadside and curbside wall areas of the operational laboratory spaces. The countertops shall serve as mounting surfaces for major items of equipment and test apparatus. Support apparatus and portable equipment will be located and stored within the cabinets during transport and when not in use. Neither the cabinetry, nor the equipment mounted upon or within it, shall block or interfere with the aisle or means of egress from any compartment. The cabinets shall be in accordance with the requirements of 3.12.

3.4.3 Undercarriage modifications. Units one and two of the MBPL shall have roadside and curbside undercarriage stowage compartments which provide for the storage of spares and equipment required for 90-day deployment. These stowage compartments shall be removable for and C-141 or larger air transport.

3.5 Components. Items of equipment for unit one of the MBPL shall be as specified in the components parts list (CPL) SC6640-97-CL-E02, and for unit two in CPL 90106. When specified (see 6.2), the items of equipment shall be furnished by the contractor and shall be located within units one and two as specified. That equipment which is countertop mounted shall be secured in such a fashion as to ensure transportability while retaining operational character.

3.6 Purging. Units one and two of the MBPL shall be purged prior to entry by energizing the evaporator coil fans within the four environmental control units (air conditioners) located on the front of each semitrailer. All other electrical power to either unit's interior systems shall be suspended during the purge cycle. Electrical componentry from the purge control to the environmental control units shall be explosion proof. During purge, dampers in the air conditioning intake air filter box shall allow only external air to be supplied to the fans. Exhaust vents shall be located low on the roadside and curbside walls to ensure evacuation of heavier-than-air gases or vapors. Purge system components shall be selected to accomplish a purge of four air exchanges in no more than 5 minutes. A purge cycle timer shall be provided on

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the exterior wall near the door and shall be adjustable from 5 minutes to 20 minutes.

3.7 Interior environment. The interior environment for units one and two of the MBPL shall be controlled during operational employment. The interior environment shall be in accordance with 3.12 except as noted herein.

3.7.1 Ambient temperature. Heating shall be provided to maintain interior dry bulb temperature above 50 °F. Air conditioning shall be provided to maintain an interior effective temperature at or below 75 °F.

3.7.2 Air supply. Ventilation air shall be introduced at a minimum of 30 sc feet per minute; approximately two-thirds shall be external air. The availability of external air during normal employment for units one and two of the MBPL shall be variable from 0 to 100 percent. The fresh air supply shall be incremental and controlled by activation of the purge and recirculation damper vents in such sequence as to provide up to 100 percent fresh air circulation in each unit.

3.7.3 Environmental controls. The environmental controls and labeling for units one and two of the MBPL shall be located on the forward roadside wall of each unit. The controls shall be in accordance with 3.12.

3.8 Utility systems. Units one and two of the MBPL shall provide self-contained water, compressed air, and vacuum utility systems. The major apparatus shall be located in the utility compartment (mechanical room) of each unit, and each service shall be distributed as necessary via copper or stainless steel tubing.

3.8.1 Water systems. The water systems in units one and two of the MBPL shall provide water to any equipment or test which requires water as well as to the sink, safety shower, and eyewash.

3.8.2 Compressed air systems. The compressed air systems in units one and two of the MBPL shall provide pressurized, oil free air at 35-85 psi to outlets (petcocks) along the roadside and curbside walls. The outlets shall be located in such a manner as to ensure the availability of compressed air to any test being performed in the laboratory.

3.8.3 Vacuum systems. The vacuum systems in units one and two of the MBPL shall supply vacuum to outlets (petcocks) along the roadside and curbside walls. The outlets shall be located in such a manner as to ensure the availability of vacuum to any test location or apparatus requiring vacuum. Vacuum supply shall be 13.5 MPa for each laboratory.

3.8.4 Color code marking. All utility piping and valving shall be color coded in accordance with MIL-STD-101, labeled and marked in accordance with 3.12.

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3.9 Electrically operated utility and testing equipment and systems. The MBPL's electrically operated utility and testing equipment and systems shall be as specified in MIL-STD-454 and shall operate as intended without malfunction or evidence of binding or overheating.

3.9.1 Power cables and power entry receptacles. A power entry receptacle shall be located on the exterior of each unit. The power entry receptacle shall be labeled "POWER ENTRY" and labeled with its voltage, phase, current and frequency characteristics. Units one and two of the MBPL shall have a 100-foot power cable stowed in one of the undercarriage storage boxes. The power distribution center will carry a 50-foot cable. The power entry receptacle connector and its mating cable shall conform to MIL-C-22992. The main power receptacle connectors and cable assemblies shall be rated waterproof in accordance with NEMA standards in the National Electric Code (NEC). Markings and labeling shall be in accordance with 3.12.

3.9.2 Circuit panels. Circuit panels shall be as specified and, when specified (see 6.2), shall be of the dead front design in accordance with W-P-115, type 1.

3.9.3 Light switch and lamp location. Light switch and lamp locations shall be in accordance with 3.12. Illuminance levels shall meet or exceed the limitations within 3.12.

3.10 Electrical wiring. All electrical wiring shall be as specified and shall meet or exceed the requirements in the National Electrical Code (NEC) of the National Fire Protection Association (NFPA). All cables and wires shall be in accordance with 3.12 and MIL-STD-454.

3.10.1 Cords. All electrical cords shall be as short as practicable while allowing for maintenance.

3.10.2 Permanent wiring. Major items of equipment which are permanently tied into the electrical system shall be equipped with "ON" and "OFF" switches as well as circuit breakers or fuses.

3.10.3 System grounding. All electrical equipment within, and each unit of, the MBPL shall be grounded in accordance with NEC and OSHA regulations and MIL-STD-1857. Ground busses and frames within units one and two of the MBPL shall be grounded to each other and to a main grounding terminal in or near the main power receptacle. The grounding terminal shall be labeled "GROUND". A ground rod shall be furnished with each MBPL unit and shall be in accordance with W-R-550, type II, class B. Grounding instructions shall be furnished on a metal instruction plate adjacent to each unit's power entry receptacle (port).

3.11 Safety equipment. The process definition of the MBPL indicates that certain tests may endanger the operating personnel if the laboratory unit does not incorporate safety devices and equipment in its design. The devices and

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equipment shall be positioned as specified to ensure the safest working conditions while still enabling the MBPL to fulfill its mission.

3.11.1 Safety equipment marking. All safety equipment shall be clearly marked in accordance with 3.12.

3.12 Human factors engineering. The MBPL shall comply with the design criteria of MIL-STD-1472. Special design emphasis shall be given, but not limited to, general requirements (4), control/display integration (5.1), visual displays (5.2), audio displays (5.3), controls (5.4), labeling (5.5), anthropometry (5.6), workspace design (5.7), environment (5.8), design for maintainability (5.9), and hazards and safety (5.13) of MIL-STD-1472, as applicable.

3.13 Noise limits. The MBPL, excluding the mobile, semitrailer mounted petroleum laboratory (unit 1), shall conform with MIL-STD-1474 requirements, with the exception of MIL-STD-1474, 5.2, 5.3, and 5.4, when tested in accordance with 4.5.2.7. Noise inside the MBPL shall not exceed MIL-STD-1474, category F, with the exception of the utility compartments which will not exceed category E, when power generation and environmental control equipment is operating under full load and laboratory equipment and accessories are operating under typically encountered conditions.

3.14 Hazardous materials. Provisions for the transportation of chemicals, compressed gases, and other hazardous materials shall comply with Department of Transportation Regulation 49CFR171-190.

3.14.1 Chemical safety. Material safety data sheets (OSHA Form 20 or DD Form 1813) in accordance with FED-STD-313 shall be provided for each chemical to be delivered with the MBPL and at the request of the contracting officer for chemicals associated with the MBPL.

3.14.2 Safety data sheet. Material safety data sheets prepared in accordance with FED-STD-313 shall be furnished for all materials which are identified as hazardous (see 6.9).

3.15 Vehicle weight classification. The contractor shall complete the Government-furnished vehicle weight classification forms and apply the vehicle weight classification number(s) to units one and two of the MBPL as specified (see 6.7).

3.16 Cushion material. The cushion material for cabinets, cupboards, drawers and shelves shall be as specified in PPP-C-1752.

3.16.1 Installation. The cushion material shall be installed in all cabinets, cabinet drawers, cupboards, and on all cabinet shelves except the overhead bookshelves (bookcases), in a form corresponding to the interior of same. The contour and depth of the cushion material shall be sufficient to protect stored equipment from impact and vibration damage. The cushion material shall be altered in such fashion as required to conform to the

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profile of the stored equipment while allowing access to, and removal of, the equipment.

3.17 Transportability. Provisions shall be made within units one and two of the MBPL to secure all equipment, supplies, chemicals, books, binders and manuals. When transported as specified herein, no damage, dislocation, breakage, cracking, or leaking shall be evident. The MBPL, fully equipped (meaning all equipment, furnishings, and supplies installed and secured for transportation), shall be transportable worldwide by the highway, water, and air modes. Transportability criteria are defined in MIL-STD-1366. Vehicles, semitrailer vans and all other materials shall not be loaded beyond their rated capacities at any time. All slinging, tiedown, and multipurpose provisions shall be labeled "LIFT" or "TIEDOWN" as appropriate, in 1-inch (2.54 cm) high letters.

3.17.1 Highway. Units one and two of the MBPL, fully equipped, shall sustain over-the-road, across curbing, and unimproved roadway transport by 5-ton truck-tractors, military design M932 or equal. When towed by the M932 or equal under any of the conditions given in 3.17 and MIL-S-46756, 3.5, units one and two of the MBPL shall experience no interference with the M932 or equal tractor.

3.17.2 Tiedown Provisions. The tiedown provisions for the DED 60kW trailer-mounted generator set, PDM, and for units one and two of the MBPL, including their bogie axles (when removed for air transport) shall conform to MIL-STD-209, class 2 or 3, type IV equipment, and to MIL-STD-1791 for equipment restraint criteria. The tiedown provisions shall satisfactorily complete the pull testing as specified without weld failure, permanent deformation cracking, loosening, or breaking of the provision or its connecting structural components.

3.17.3 Slinging Provisions. The slinging provisions for the DED 60kW trailer-mounted generator set, PDM, and for units one and two of the MBPL, including their bogie axles (when removed for air transport) shall conform to MIL-STD-209 class 1 or 3, type II. The provisions shall enable the complete MBPL to be lifted in the normal operating position. Any structure of an item (such as the corners where the top and side panels of the semitrailer vans meet) that creates interference with a lifting sling will have the strength to withstand contact with the sling during pull testing without permanent deformation to any part of the item, and the contact will not adversely affect the sling. Slinging provisions may also be used as tiedown provisions when such provisions meet the requirements specified in 3.17.2.

3.17.4 Air. Units one and two of the MBPL, fully equipped, shall be transportable in a reduced configuration by C-141 or larger aircraft. All items removed for transport to include the tandem bogie axle and stowage compartments shall be capable of being transported with their respective semitrailer van. Units one and two of the MBPL, fully equipped, shall be roll-on/rolloff (RORO) transportable aboard C-5 aircraft without disassembly.

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Criteria for transport in U.S. Air Force cargo aircraft are defined in MIL-STD-1791.

3.17.5 Water. Units one and two of the MBPL, without disassembly, shall be capable of transport aboard breakbulk (general cargo), RORO, and barge-carrying (LASH and SEABEE) ships. Units one and two of the MBPL, when towed by the M932 or equal, shall be able to approach, negotiate, and depart a 15 degree angle ramp without disassembly, without undercarriage interference, and without loss of tire to road surface contact.

3.18 Power distribution module.

3.18.1 Description. The power distribution module shall be four-soldier portable, comply with the transportability requirements in 3.17 and shall have lifting and slinging provisions for internal and external helicopter transport. The unit shall be water resistant (shall be sealed to prevent internal damage and have a corrosive resistant coating (paint) on exterior surface) and shall operate in temperatures from -25 °F to 120 °F. The high and low temperature testing for the PDM is outlined in 4.8. The module will provide the means by which power available from a single source (i.e., generator, power line, etc.) can be split and routed to units one and two of the MBPL. The module shall contain such circuitry as is required to protect a 60 KW diesel engine driven generator set from overload.

3.18.2 Design. The power distribution module shall be designed and fabricated within the restraints and parameters herein. The design shall be such as to limit or prevent conditions hazardous to personnel or equipment. The weight of the power distribution module, with the cable removed, shall not exceed 147 pounds. In addition, the components, and installation thereof, of the electrical system shall be in accordance with the National Electric Code (NEC), NFPA-70, and MIL-STD-454. Unit grounding shall be in accordance with MIL-STD-1857. No deviation from the prescribed limitations or equipment specifications is permissible without prior approval of the contracting officer. Any data (e.g., shop and fabrication drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a supplier to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or designated representative (see figures 1 and 2).

3.18.3 Operation. The power distribution module shall provide a means by which power, supplied by a military designed generator or commercial land line, can be split and apportioned to units one and two of the MBPL. Terminations shall be provided for three 5-wire cables. The input power cable, originating at the power source, shall be transported by, and permanently attached to, the power distribution module. The terminals, to which the cables supplying power to units one and two of the MBPL shall be attached during operation, shall be supplied apportioned power via power contractors. The contractor for unit one of the MBPL shall have a rated minimum of 208 amperes and the contractor for unit two shall have a rated minimum of 100 amperes. Each contact shall be controlled by separate push-

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off/pull-on switches located on the power distribution module control panel. In addition, the contractor control circuitry shall incorporate the capability to detect an overcurrent condition of 208 amperes load on the main power cable input and automatically trip off the contractor controlling MBPL unit two's power supply. A battle override and separate reset switch shall enable manual power resupply or override capability at the commander's discretion. Terminals conforming to MS39347 shall provide the interface connection for each unit of the MBPL. Interlocks shall be provided on all access doors and covers to provide personnel protection against electrical shock.

3.18.4 Instrumentation. The design of the power distribution module control panel shall incorporate a frequency meter, a voltmeter and two elapsed-time meters, one for each laboratory unit supplied. In addition, circuitry shall be provided for an "out-of-phase" detector and a panel-mounted indicator light. The lights shall be switch activated. Meter and switch illumination shall be provided by lights meeting the following requirements.

a. Light sources shall consist of incandescent bulbs with filters and dimmers or light emitting diodes (LEDS).
b. Ninety-five percent of the light energy emitted by each light source shall be at wavelengths below 700 nanometers.
c. Incandescent light sources shall be dimmable to 0.05 foot-lamberts (FL) or less.
d. All light sources shall be visible in bright sunlight.
e. All light sources shall be controlled by a single on/off blackout switch, labeled "BLACKOUT SWITCH".

3.18.5 Electrical components. Electrical components used in the design and fabrication of the power distribution module shall be UL approved and conform to NEC and OSHA standards. All electric and electronic components shall be fabricated from virgin raw material. The electrical components used in the fabrication of the module shall be as specified on figures 1, 2, 3, and 4 or equal.

3.18.6 Power entry cable. The power entry cable shall be 3 phase, 600 volt, 8 wire (4 wires #4/0 and 4 ground wire #4 AWG) cable, 50 feet in length, minimum. The cable shall be permanently attached to the power distribution module. Provisions shall be made to store and transport the cable on the power distribution module frame. The generator end of the cable shall be prepared in such fashion as to interface with a military design generator.

3.18.7 Construction. The material used in the construction of fabricated components, except fastening devices, shall be aluminum alloy. The use of rivets and riveting shall be avoided. All fasteners shall be manufactured from nonsparking, nonferrous materials. When structural shapes and tubing are used in the fabrication of power distribution module components, seamless shapes and tubing shall be used. All tubing ends shall be closed. All exterior surfaces, except gauges, meters, lights and switches shall be treated and painted in accordance with MIL-T-704, type F or G, as applicable.

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3.18.7.1 Support frame and cable carrier. The power distribution module support frame shall be designed to provide adequate support, during transport, for the main power cable and the electrical component chassis assembly. The support frame shall be 30.5 inches high, 45.75 inches long, and 25.75 inches deep, and shall be constructed from 3-inch by 1-inch rectangular tubing with 0.125-inch wall. The tubing shall be welded "side-on-side" to form a 2-inch by 3-inch section, the 3-inch side always located in the vertical plane (see figure 1). Aluminum sheet, 0.25 inch thick, formed into a 16-inch diameter half-circular shape, shall be welded to the vertical frame members, thus creating the cable carrier. The cable carrier shall be 16 inches in depth. Two 1.75-inch by 3-inch, 0.125-inch wall, rectangular tubes shall be positioned at the corners of the side of the frame from which the cable is accessed. Each tube shall be pivoted at its lower end and secured at the top, during transport, by removable pins. The design shall be such that the pivots and pins are in shear.

3.18.7.2 Electrical component chassis. The electrical component chassis shall be so designed as to contain and protect the high voltage and control devices required by the power distribution module during operation (see figure 1). The chassis shall be comprised of a chassis base, an environmental cover, a contractor support frame, and an insulating stage. The chassis base shall be bolted to the power distribution module support frame and incorporate in its design a recessed control panel and means by which the main power cable and unit one and unit two's power cables may enter the module while maintaining its environmental integrity. The environmental cover shall incorporate in its design an access panel, or door, enabling the operating personnel to access the insulating stage and the power output terminals mounted within it. The environmental cover shall be fastened securely to the chassis base. A silicone gasket shall provide a seal ensuring the environmental integrity of the chassis assembly. The contact support frame will be positioned within, and fastened to, the chassis base and will provide the elevated surface on which the majority of the electrical components shall be mounted. The insulating stage shall be fabricated of plastic, or similar, nonconductive material. The stage will be so designed as to prevent personnel from contacting any electrical component other than the power output terminals which shall be mounted in it. The electrical component chassis shall be 48 inches long, 26 inches deep and approximately 13 inches high, excluding the cable connections. The chassis base, environmental cover, and contactor support frame shall be fabricated from 0.125 thick sheet and shall be of bent and welded construction.

3.19 Identification marking. The MBPL and its integral units and their component parts shall be identified in accordance with MIL-STD-130. The marking shall be applied to each unit of the MBPL on plates conforming to MIL-P-514, type I, style 1, composition C, of type I, grade A, class 1 material. Plates shall be attached by screws, bolts, or rivets in a conspicuous location.

3.20 Shipping data plates. The shipping data plates shall conform to MIL-P-514, type III, composition C, of type I, grade A, class 1 material and,

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in addition, shall show the silhouette of each unit of the MBPL in transport position indicating the center of gravity and the location and capacity of the lifting and tiedown attachments. The plate shall be attached by screws, bolts, or rivets in a conspicuous location on the exterior of each unit.

3.21 Instruction plates. Each unit of the MBPL shall be equipped with instruction plates or diagrams, including warnings and cautions, describing any special or important procedures to be followed in the deployment, operation, or servicing of the units comprising the MBPL in accordance with 3.12. Instruction plates shall conform to MIL-P-514, type III, composition C, of type I, grade A, class 1 material. Plates shall be attached by screws, bolts, or rivets in a conspicuous location and in close proximity to the equipment or item(s) of concern. Instructions shall be clear and concise, summarizing steps required for the procedure being described.

3.22 Treatment and painting. The portions of the units comprising the MBPL normally painted internally and externally shall be cleaned, treated, and painted in accordance with MIL-T-704, type F or G, as applicable. When specified (see 6.2), the units shall be overcoated in accordance with Government-furnished camouflage patterns with MIL-C-46168 or MIL-C-53039 top coat of the colors specified in camouflage patterns.

3.23 Government-furnished property. When specified (see 6.2 and 6.5), the following property in the quantities indicated will be furnished by the Government:

Item no.	Description	Identification	Qty per unit one	Qty per unit two
1	Truck Tractor, Military Design, 5-Ton, 6x6, M818 or Equal.	2320-01-835-8326	1	1
2	Semitrailer, Van, M822E1.	2230-00-162-0642	1	1
3	Environmental Control Unit (ECU).	4120-00-168-1781	4	4
4	Fire Extinguisher.	4210-01-087-4989	3	3
5	Manual, ECU Parts List.	TM 5-4120-295-25P1	1	1
6	Manual, ECU Operation and Maintenance.	TM 5-4120-295-15	1	1
7	Manual, Petroleum Handling Equipment and Operations.	TM 10-1101	-	1
8	Manual, Inspecting and Testing Petroleum Products.	TM 10-70	1	1
9	Manual, Operating and Maintenance, for the Laboratory, Petroleum, Semitrailer Mounted.	FM 10-21	-	1
10	Manual, Operating and Maintenance, for the Laboratory, Petroleum, Semitrailer Mounted.	TM 5-6640-212-14	1	-

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Item no.	Description	Identification	Qty per unit one	Qty per unit two
11	Operator, Organization, Direct and General Support Maintenance Manual.	TM 9-2330-271-14	-	1
12	Coal and Petroleum Products Supply and Management Activities.	AR 703-1	-	1
13	ASTM Standards, Petroleum Products and Lubricants, Vol. 0.501.		1	1
14	ASTM Standards, Petroleum Products and Lubricants, Vol. 0.502.		1	1
15	ASTM Standards, Petroleum Products and Lubricants, Vol. 0.503.		1	1
16	Chemical Engineers Handbook.	7610-00-250-6633	1	1
17	Chemistry and Physics Handbook.	7610-00-233-9577	1	1
18	ASTM Viscosity Index.	STP-168 or D 539	1	1
19	ASTM Viscosity Tables.	STP-43-B	1	1
20	Chemical Index.		1	1
21	Chemical Technology of Petroleum and Its Products.		1	1
22	Lubricant, Liquid Fuel and Related Products, Methods of Testing.	FED-STD-791.	1	1
23	Quality Surveillance Handbook for Fuels, Lubricants and Related Products.	MIL-HDBK-200.	1	1
24	National Petroleum Correction Tables.		1	1
25	Reference List of Specifications and Standards for Petroleum and Related Products.		1	1
26	Significance of ASTM Tests for Petroleum Products.		1	1
27	TB 34-9-25 Symbols for the Identification of Fuel, Lubricants, and Allied Products - NATO.		1	1
28	Technical Methods of Analysis.		1	1
29	Manual, Aircraft Refueling.	FM 10-68	1	-
30	Manual, Petroleum Supply Point.	FM 10-69	1	1
31	Supply Catalog, Laboratory, Petroleum Semitrailer-Mounted.	SC 6640-97-CL-E02	1	-
32	Supply Catalog, Laboratory, Petroleum, Semitrailer-Mounted, Supplemental.		-	1
33	Tool Kit, General Mechanics.	5180-00-177-7033	1	1
34	Pattern Drawings.	Camouflage Patterns	1	1
35	ACS Chemicals.	MIL-STD-1218	1	1
36	Sign Kit, Vehicle Class.	MIL-S-40626	1	1
37	Distillate Fuel Contamination, Storage, and Handling.	ASTM STP 1005	1	1

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Item no.	Description	Identification	Qty per unit one	two
38	Guide for the Selection of Lubricants, Functional Fluids, Preservatives, and Specialty Products for Use in Ground Equipment Systems.	MIL-HDBK-113	1	1
39	Mobility Fuels User Handbook.	MIL-HDBK-114	1	1

3.24 Government loaned property. When specified (see 6.2), the following property in the quantity indicated will be loaned by the Government (see 6.6).

Item No.	Description	Quantity
1	M-52 truck tractor NSN 2320-01-835-8326 or equal	1

3.25 Workmanship.

3.25.1 Fabrication. Metal used in the fabrication of the MBPL shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the metal to a value less than intended by the design. Manufacturing processes shall be accomplished neatly and accurately. All bends shall be made by controlled means to ensure uniformity of size and shape.

3.25.2 Welding. The surfaces of parts to be welded shall be free from rust, oxides, scale, paint, grease, or other foreign matter. Weldment shall be of homogeneous appearance and form without spattering or other irregularities, complying with the size, configuration, and other dimensional requirements to develop the full strength of the parts joined by the welds. No cracks of any type are permitted. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.25.3 Welders. Before assigning any welder or welding operator to manual welding work covered by this specification, the contractor shall obtain certification that the welder and welding operator has passed qualification tests as prescribed by either the following listed codes for the materials joined and the type of welding operation to be performed and that such qualification is effective as defined by the particular code.

AWS D1.1, Structural Welding Code - Steel
AWS D1.2, Structural Welding Code - Aluminum
ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualification

Contractors who only make horizontal welds need not qualify welders for "all position welding." The contractor is responsible for determining that

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automatic welding equipment operators are capable of producing quality welds in accordance with AWS and ASME codes. In the event of evidence of poor welds, the Government reserves the right to require retesting of any welder or welding operator. The test results shall be made available for review by the contracting officer or the contracting officer's representative.

3.25.4 Welding practices.

a. Tack welds shall be subject to the same quality requirements as final welds.
b. Work shall be positioned for horizontal welding whenever practicable.
c. Procedures and sequences shall be such that distortion and shrinkage will be held to a minimum. When straightening is required, caution shall be exercised to ensure that the straightening process does not weaken the part, the assembly of which it is a component, or the unit.
d. Preheat of materials being welded and maximum interpass temperatures during welding shall be in accordance with the suppliers' recommendations.

3.25.5 Weld quality.

a. The contractor shall be responsible for determining the inspections needed to ensure that when the welded parts are assembled together to make an assembly, the unit shall conform to the inspection requirements specified in section 4.
b. All weldments shall be free of slag, flux, weld spatter, and other impurities detrimental to both the appearance and strength of the weldment.
c. Undercut in weldments shall be held to a minimum, but shall not be more than 0.03125 inch deep or 5 percent of the thickness of the thinner material, whichever is less.
d. Welds shall be free from all cracks in adjacent base material and weld metal and from other discontinuities such as lack of fusion with base metal or between weld passes, incomplete penetration or lack of prescribed fit.
e. The weld quality shall be in accordance with AWS D1.1 or AWS D1.2, as appropriate.

3.25.6 Brazing. Brazing of steels, copper alloys and nickel alloys shall conform to MIL-B-7883.

3.25.7 Bolted connections. Bolt holes shall be accurately formed and shall have the burrs removed. Washers and lockwashers shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall have full thread engagement. All threads shall be in accordance with FED-STD-H28.

3.25.8 Rivet connections. Rivets and installation thereof shall be in accordance with drawing 13217E1061.

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3.25.9 Machine work. Tolerances and gauges for metal fits shall conform to the standards of the industry.

3.26 Maintainability. The MBPL shall permit routine service and maintenance under military field conditions. Components which require periodic servicing as a matter of routine maintenance shall be readily accessible.

3.26.1 Maintenance support. Assemblages or support elements shall be adequate to perform the intended function of assisting or conducting maintenance operations when tested as specified in 4.5.2.10.

3.27 Environmental.

3.27.1 Operating temperature. The MBPL shall operate in ambient air temperatures ranging from -25 °F to +120 °F. Special environmental equipment shall be supplied as required.

3.27.2 Storage temperature. The MBPL shall not be damaged by storage at ambient temperatures from -28 °F to +160 °F.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility of compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for ensuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

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4.2 Classification of inspection. Inspection shall be classified as follows:

a. First article MBPL inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.7).

4.3 First article MBPL.

4.3.1 Examination. The first article MBPL shall be examined for the defects specified in table I as applicable. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article MBPL shall be tested as specified in 4.5.2.1 through 4.6.1. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Examination. The MBPL shall be examined for the defects specified in table I as applicable. Presence of one or more defects shall be cause for rejection.

4.4.2 Tests. Units one and two of the MBPL shall be tested as specified in 4.5.2.1, 4.5.2.5, 4.5.2.6 and 4.6.1. Failure of any test shall be cause for rejection.

4.5 Inspection schedule.

4.5.1 Examination. Examination shall be in accordance with table I.

TABLE I. Examination

Number	Defects	Requirement paragraph
101.	Assembly incomplete or incorrect.	3.1
102.	Materials not as specified.	3.3
103.	Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments.	3.3.1
104.	Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.	3.3.2
105.	Contractor does not have documentation available for identification of material, material finishes, or treatment.	3.3.3
106.	Used, rebuilt or remanufactured components, pieces, or parts incorporated in the MBPL.	3.3.4
107.	Design detail not as specified.	3.4

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TABLE I. Examination - continued.

Number	Defects	Requirement paragraph
108.	Semitrailer modifications incomplete or incorrect.	3.4.1
109.	Cabinetry not as specified.	3.4.2
110.	Undercarriage modifications not as specified. specified.	3.4.3
111.	Parts and components missing or not as specified.	3.5
112.	Van purging design not as specified.	3.6
113.	Environment parameters not as specified.	3.7
114.	Ambient temperature not as specified.	3.7.1
115.	Ducting not as specified.	3.7.2
116.	Utility systems not as specified.	3.8
117.	Piping not as specified.	3.8.1
118.	Compressed air systems not as specified.	3.8.2
119.	Vacuum system not as specified.	3.8.3
120.	Piping color code marking not as specified.	3.8.4
121.	Electrically operated utility and test equipment and systems not as specified.	3.9
122.	Power cables and power entry receptacles not as specified.	3.9.1
123.	Dedicated computer power circuit not as specified.	3.9.2
124.	Illumination not as specified.	3.9.3
125.	Electrical wiring not as specified.	3.10
126.	Electrical cords not as specified.	3.10.1
127.	Permanent wiring not as specified.	3.10.2
128.	System grounding not as specified.	3.10.3
129.	Safety equipment not as specified or missing.	3.11
130.	Placards and labels missing or incorrect.	3.12
131.	Human factors engineering not as specified.	3.12
132.	Provision for the transportation of hazardous material not as specified.	3.14
133.	Material safety data sheets missing or not prepared in accordance with FED-STD-313.	3.14
134.	Vehicle weight classification missing or not as specified.	3.15
135.	Cushion material not as specified.	3.16
136.	Cushion material not installed as specified.	3.16.1
137.	Identification marking and identification plates not as specified or missing.	3.19
138.	Shipping data plate not as specified or missing.	3.20
139.	Instruction plates not as specified or missing.	3.21
140.	Treatment and painting not as specified.	3.22
141.	Paint color not as specified.	3.22

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TABLE I. Examination - continued.

Number	Defects	Requirement paragraph
142.	Workmanship not as specified.	3.25
Power Distribution Module (PDM)		
143.	Transportability not as specified.	3.17-3.17.5
144.	Assembly incomplete or incorrect.	3.18
145.	Design detail not as specified	3.18.2
146.	Instrumentation not as specified.	3.18.4
147.	Electrical components not as specified.	3.18.5
148.	Power cable not as specified.	3.18.6
149.	Construction not as specified.	3.18.7
150.	Support frame and cable carrier not as specified.	3.18.7.1
151.	Electrical component chassis not as specified.	3.18.7.2

4.5.2 Tests.

4.5.2.1 Test conditions. Tests shall be performed at the climatic conditions existing at the place of test. The Government shall pre-approve all test courses. Each unit of the MBPL shall operate as specified herein without maintenance other than the contractor's recommended normal scheduled maintenance as established by a maintenance schedule prepared and submitted by the contractor prior to test.

4.5.2.2 Road. Units one and two of the MBPL, with all equipment, furnishings and supplies installed, shall be coupled with a Government-loaned M818 truck tractor, or equal, and towed over a hard-surfaced road for a minimum of 200 miles at sustained speeds up to at least 50 miles per hour. A circular test track shall not be used for this test. In addition, the combination shall negotiate at least 50 miles of trails (secondary unimproved roadway) at safe speeds (not less than 5 mph). Nonconformance to 3.17 shall constitute failure of this test.

4.5.2.3 Across curbing. Units one and two, fully loaded and equipped, of the MBPL shall be towed by the M-52 truck tractor (or equal) over a section of concrete, asphalt, or solid wood curbing at not less than 5 miles per hour. The dimensions of the curbing shall be 6 inches, ±0.50 inch, high by 6 inches, ±0.50 inch, wide. The top corners of the curbing coming into contact with the wheels of the tractor and the MBPL shall be rounded to a 1-inch, ±0.125 inch, radius. The curbing shall be anchored to a hard-surfaced road and shall be stationary during the test. The full height of the curbing shall protrude above the surface of the road. All wheels of the tractor and MBPL shall pass

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over the curbing at a 45-degree, ±5 degrees, angle to the centerline of the curbing with the road side of the tractor and MBPL. This test shall be repeated passing all wheels of the tractor and MBPL over the curbing at a 45-degree, ±5 degrees, angle to the centerline of the curbing with the curb side of the tractor and MBPL. Nonconformance to 3.17 shall constitute failure of this test.

4.5.2.4 Slinging and tiedown provision pull test. When specified (see 6.2), to determine conformance to 3.17.2 and 3.17.3, the lifting and slinging provisions for the 60 kW DED trailer-mounted generator set, and for units one and two of the MBPL, including their bogie axles (when removed for air transport), shall be tested in accordance with MIL-STD-209. Test results that show that the provisions and the connecting structural members do not meet the requirements of MIL-STD-209 shall constitute failure of this test.

4.5.2.5 Pressure testing of utility system. The MBPL service utility systems shall be tested as indicated:

System	Pressure	Medium	Duration*	Examination
Water	90 psig	Water	30	Visual
Drainage	Gravity	Water	10	Visual
Air	120 psig	Air	30	Soap solution
Vacuum	13.5 MPa	Negative pressure	30	Manometer

*Not less than time indicated in minutes

The above testing and examination shall be conducted for a total of two times. After the first test, the pressures shall be allowed to return to ambient before pressurizing for the second test. Any evidence of leakage shall constitute failure of this test.

4.5.2.6 Functional testing of electrically operated systems and equipment. The electrically operated equipment and electrical systems shall be electrically energized and operated. With power being supplied by the 60kW generator, the power distribution module shall split and apportion the power to units 1 and 2 of the MBPL.

a. Time-delay purging system.
b. Lighting blackout system.
c. Utility equipment.
d. Testing equipment.
e. 208/115-volt ac electrical system.
f. PDM

Nonconformance to 3.9 or 3.18 shall constitute failure of this test.

4.5.2.7 Noise level test. Noise levels shall be measured in accordance with MIL-STD-1474 requirements and reported in the format indicated by

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MIL-STD-1474, figure 11. As a minimum: noise levels shall be measured when power generation and environmental control equipment is operating under full load and laboratory equipment and accessories are operating under typically encountered worst case conditions. MIL-STD-1474 contours shall be taken at not fewer than 12 equal arc increments; one increment shall include data from the noisiest position. Additionally, the noise level at the typical operating or working position(s) shall be provided as dB(A) level. Failure to comply with the requirements of 3.13 and MIL-STD-1474 shall constitute failure of this test.

4.5.2.8 Transport by water test. Units one and two of the MBPL shall be evaluated for conformance with 3.17.5. To determine conformance to 3.17.5, units one and two of the MBPL, without disassembly, towed by the M932 or equal, shall approach, negotiate, and depart a 15 degree angle ramp. Inability to be transported by breakbulk, RORO, and barge-carrying ships, or inability to approach, negotiate, and depart a 15 degree ramp angle, when towed by the M932 or equal, without disassembly, undercarriage interference, and loss of tire to road surface contact shall constitute failure of this test.

4.5.2.9 Fixed wing aircraft loading test. The MBPL shall be inspected and calculations provided to show that it meets the criteria of MIL-STD-1791. Unless otherwise specified (see 6.2), the contractor will prepare for loading, load, and secure unit one or unit two of the MBPL in a C-130 aircraft in conformance with 3.17.4. Nonconformance to 3.17.4 or the criteria in MIL-STD-1791 shall constitute a failure of this requirement.

4.5.2.10 Maintainability. Maintenance operations required during testing shall be accomplished to conform to 3.26. Failure of the support elements to maintain the MBPL as specified shall constitute failure of this test.

4.6 Power distribution module.

4.6.1 Operational. The power distribution module (PDM) shall operate as designated in 3.18.3. To determine conformance with 3.18.3, the PDM will be hooked into both units with the 60 KW generator and various test equipment operating. With power being supplied by the designated generator, the PDM shall split and apportion the power to units one and two of the MBPL.

4.7 Environmental testing.

4.7.1 High temperature. The MBPL system shall be tested in a high temperature environment in accordance with MIL-STD-810, method 501, procedures I and II. Temperature sensors shall be placed in all compartments of unit one and unit two (utility, main, computer) and the PDM. The sensors shall be placed at locations away from vents and any direct air flow. Test temperatures shall be as stated in 3.27. Test duration shall be seven cycles for procedures I and constant temperature exposure for II. The system shall be operated until all recorded temperatures have stabilized or a minimum of 4 hours. Interior operating temperatures must be between 50 °F and 75 °F for a

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minimum of 4 hours. Any damage, leakage or failure of the system to operate as required in this specification shall constitute failure of this test.

4.7.2 Low temperature. The MBPL system shall be tested in a low temperature environment in accordance with MIL-STD-810, method 502, procedures I and II. Temperature sensors shall be located in all compartments (utility, main, computer room) of unit one, unit two and the PDM. Sensors shall be placed in locations away from vents and any direct air flow. Test temperatures shall be as stated in 3.27. Interior operating temperatures must be maintained between 50 °F and 75 °F for a minimum of 4 hours. Test duration for storage is 4 hours and a minimum of 4 hours for operation until recorded temperatures stabilize. Any damage, leakage, or failure of the system to operate as required in this specification shall constitute failure of this test.

4.8 Inspection of packaging. The packing and marking of each MBPL shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

152. The MBPL not preserved as specified for level A or C (see 5.1).
153. Equipment, supplies, accessories, overpack boxes not positioned and secured as specified for level A or C (see 5.1.1.2).
154. Overpack boxes, if supplied, not as specified for level A or C (see 5.1.1.2).
155. Drive belt tension not loosened as specified for level A (see 5.1.1.2).
156. Pulley grooves not coated with primer as specified for level A (see 5.1.1.2).
157. Ferrous metal surfaces requiring a preservative and not coated as specified for level A (see 5.1.1.2).
158. Specific types of items not preserved as specified for level A (see 5.1.1.2).
159. Fluorescent lamps not secured as specified for level A or C (see 5.1.1.2).
160. Exterior openings into air conditioner not closed as specified for level A (see 5.1.1.3).
161. Doors, blackout covers, and weather flaps not closed and secured as specified for level A or C (see 5.1.1.4).
162. Marking incomplete, illegible, incorrect, or missing for level A or C (see 5.3).

5. PACKAGING

5.1 Preservation. Preservation of the laboratory trailer and all components of the MBPL shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Trailer. The trailer shall be preserved in accordance with the applicable requirements of MIL-V-62038 for level A towed equipment.

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5.1.1.2 Laboratory components. All equipment and accessories shall be located and secured in their designated locations. Any components and accessories having electrical components such as motors, switches, etc., shall be preserved in accordance with MIL-P-116, method IIC. Any components which will not fit into designated cabinet space because of increased size due to preservation shall be packed in one or more boxes conforming to PPP-B-601, domestic type. Components shall be secured within the box(es) with blocking and bracing to prevent movement of contents. Exposed, unprotected ferrous metal surfaces of mounted equipment shall be coated with appropriate preservatives in accordance with MIL-P-116. All equipment driven by belts shall have the belt tension loosened and the pulley grooves coated with primer conforming to TT-P-664. Fluorescent lamps shall be secured in position by taping over the slot of the holders with 1-inch wide tape conforming to PPP-T-60, type IV. Manuals and books shall be preserved in accordance with MIL-P-116, method 1C-3. The refrigerator door shall be propped and secured in the open position to prevent permanent set of the door gasket. Overpack boxes, if included, shall be secured inside the trailer in a manner which will prevent box movement in any direction.

5.1.1.3 Air conditioning unit. Exterior openings into the air conditioning unit not provided with covers shall be closed with waterproof barrier material conforming to PPP-B-1055, class H-5, secured in place with waterproof tape. A warning tag shall be prominently affixed to the air conditioning controls, and shall caution against operating the unit without all of the protective storage or transport material.

5.1.1.4 Door and exterior openings. The doors, blackout covers, and weather flaps shall be closed and secured. All inlets, outlets, joints around doors, covers, and flaps shall be sealed with minimum 2-inch wide tape conforming to PPP-T-60, type IV. Any openings too large to be sealed with tape shall be covered with waterproof barrier material secured in place with waterproof tape.

5.1.2 Level C. The trailers shall require no preservation. All equipment, supplies, accessories, and overpack boxes shall be positioned and secured as specified for level A. Fluorescent lamps shall be secured in place with the tape as specified for level A. Doors, blackout covers, and weather flaps shall be closed and secured. Additional preservation shall be applied, as deemed necessary by the supplier, to afford protection against deterioration and damage from the supplier to the initial destination.

5.2 Packing. Packing shall be level A or C, as specified (see 6.2).

5.2.1 Level A. The MBPL preserved as specified in 5.1 shall not require additional packing.

5.2.2 Level C. The MBPL preserved as specified in 5.1 shall not require additional packing, except doors shall be equipped with tamperproof seals to assure carrier acceptance at the lowest ratings of the carrier.

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5.3 Marking. In addition to any caution and deprocessing marking, interior packs and units shall be marked in accordance with MIL-STD-129 and as specified herein. Special marking requirements applicable to unpacked vehicles shall apply. Tool, spare parts, and overpack boxes shall contain a list of items contained therein.

5.3.1 Humping precaution. Large signs containing the words "DO NOT HUMP" shall be secured in conspicuous places on each side of the MBPL.

5.4 Shipping precautions.

5.4.1 Highway towaway shipment. Prior to highway shipment, no processing of the running gears in units one and two shall be performed that would impair necessary operation or safety during transit. Towaway rules for shipment by driveaway shall be in accordance with DOT Motor Carrier Safety Regulation, 49CFR393. When vehicles are to be placed in storage or shipped overseas, processing of unit one and two's running gear not accomplished prior to highway shipment shall be performed at an intermediate designation.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The MBPL and sustaining equipment covered by this specification are to be used by Armed Forces for testing petroleum and petroleum products under tactical military operations in a theater of operations.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. Time frame required for submission of first article model (see 3.2).
d. When items of equipment shall be furnished by the contractor (see 3.5).
e. When circuit panels will be of the dead front design (see 3.9.2)
f. Weight classification numbers and instructions required (see 3.15 and 6.7).
g. When overcoat in accordance with camouflage patterns is not required (see 3.22).
h. When Government-furnished property is required (see 3.23 and 6.5).
i. When Government-loaned property is required (see 3.24 and 6.6).
j. When MIL-STD-209 provision pull testing is required for Government-furnished equipment (see 4.5.2.4).
k. When an aircraft test loading is not required for a semitrailer (see 4.5.2.9).
l. Level of preservation and packing required (see 5.1 and 5.2).

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m. When a transportability report is required (see 6.11).

6.3 First article. When a first article inspection is required, the item(s) should be a first produced MBPL. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserve the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Hazardous materials. Hazardous materials are listed in DOT Regulation 49CFR171 (see 3.14).

6.5 Government-furnished property. The contracting officer should arrange to furnish the property specified in 3.23, as applicable.

6.6 Government-loaned property. The contracting officer should arrange to loan the property specified in 3.24.

6.7 Vehicle weight classification. The contracting officer should arrange to furnish vehicle classification forms to the contractor and request the return of the completed forms to the Government. The information thus obtained should be utilized to determine the weight classification number(s) for the MBPL. The contractor should then be requested by the contracting officer to apply the weight classification number on the front of unit one and unit two of the MBPL on the curb side. The use of a Government-furnished sign kit conforming to MIL-S-40626 should be utilized where applicable.

6.8 Definition. Where an item is specified by trade name and is followed by the words "or equal" or "or equivalent," it is intended to indicate a supplier satisfactory to the Government and a quality level that must be met or surpassed by any supplier of an "or equal" or "or equivalent" item. In order to be considered equal or equivalent, a piece or item must be dimensionally and functionally interchangeable with the piece or item it is to replace.

6.9 Material safety data sheets. Contracting officers will identify those activities requiring copies of completed material safety data sheets prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in appendix B of FED-STD-313.

6.10 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DOD 5010.12-L, Acquisition Management Systems and Data

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Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.11 Transportability report. When specified (see 6.2), the contracting officer should arrange for the contractor to furnish a transportability report in accordance with AR 70-47 and S.O.P. 70-38 for the 60KW DED trailer mounted generator set, 1 and 2 of MBPL, 95 days prior to delivery of the first article.

6.12 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of the MBPL.

6.13 Subject term (key word) listing.

Electrical Distribution
Fuel Analysis
Fuel Testing
Module, Power Distribution
Testing, Petroleum

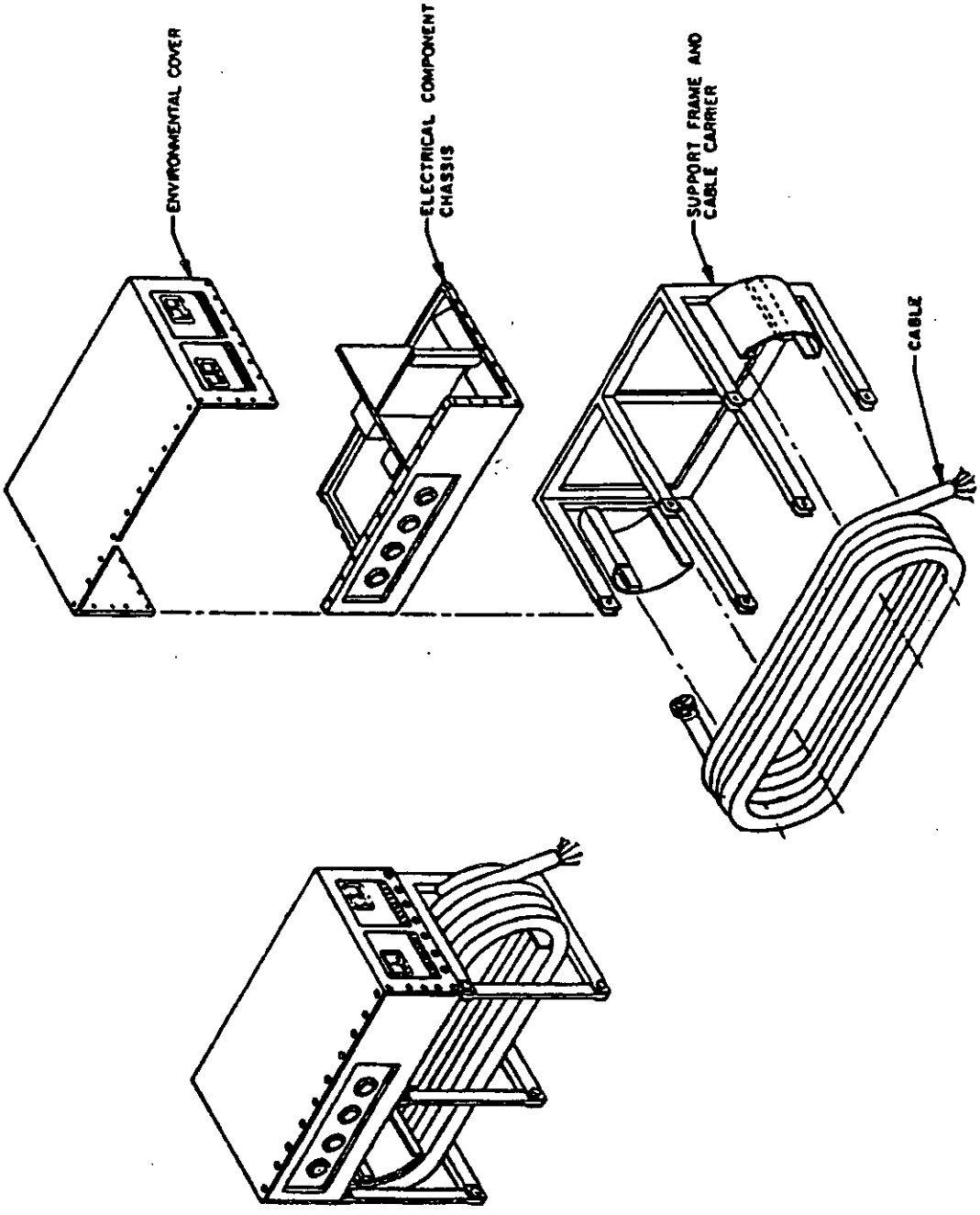
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Army - ME

Preparing activity:
Army - ME

Review activities:
Army - MD

Project 6640-A610

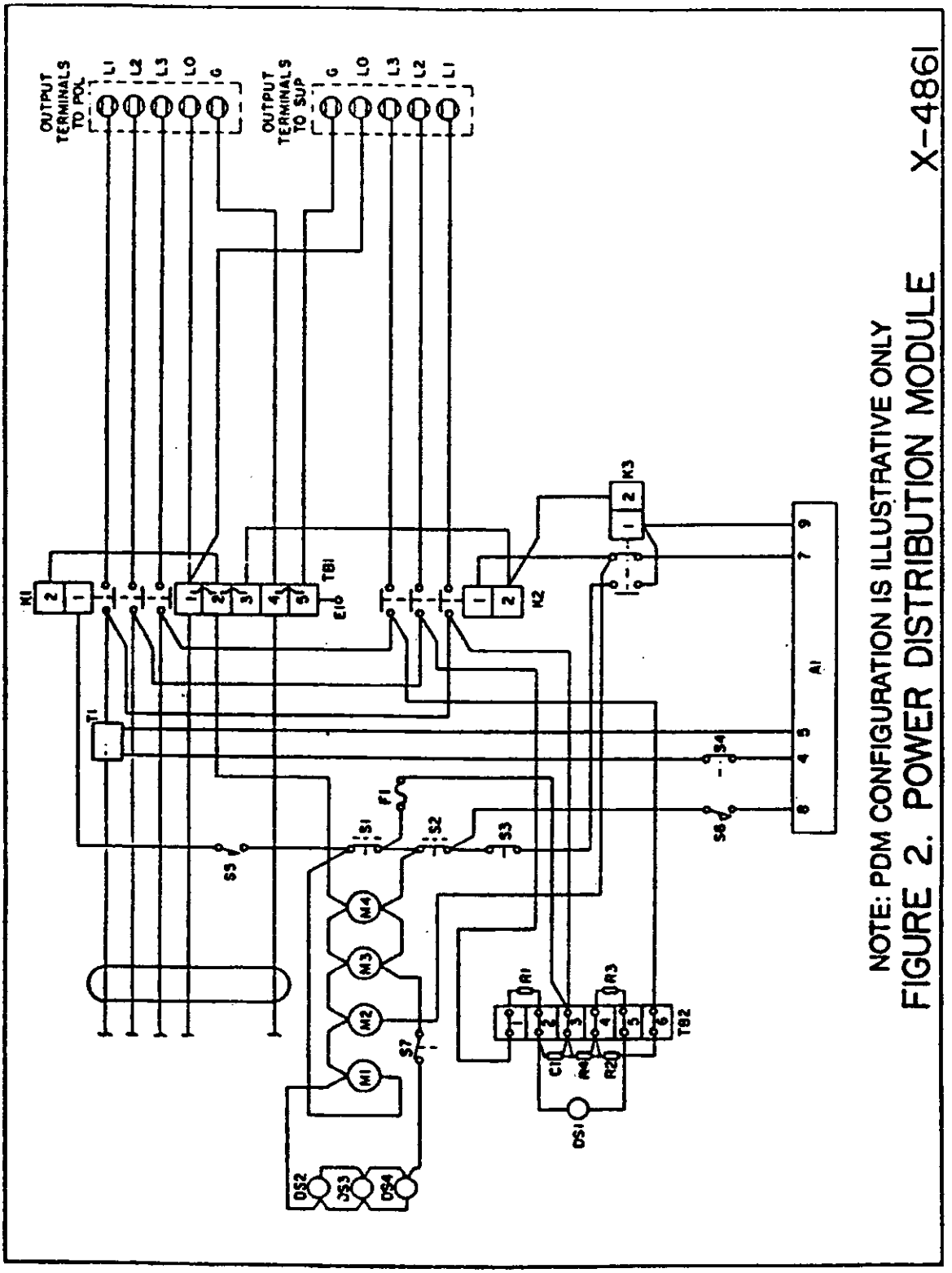
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NOTE: PDM CONFIGURATION IS ILLUSTRATIVE ONLY

FIGURE 1. POWER DISTRIBUTION MODULE

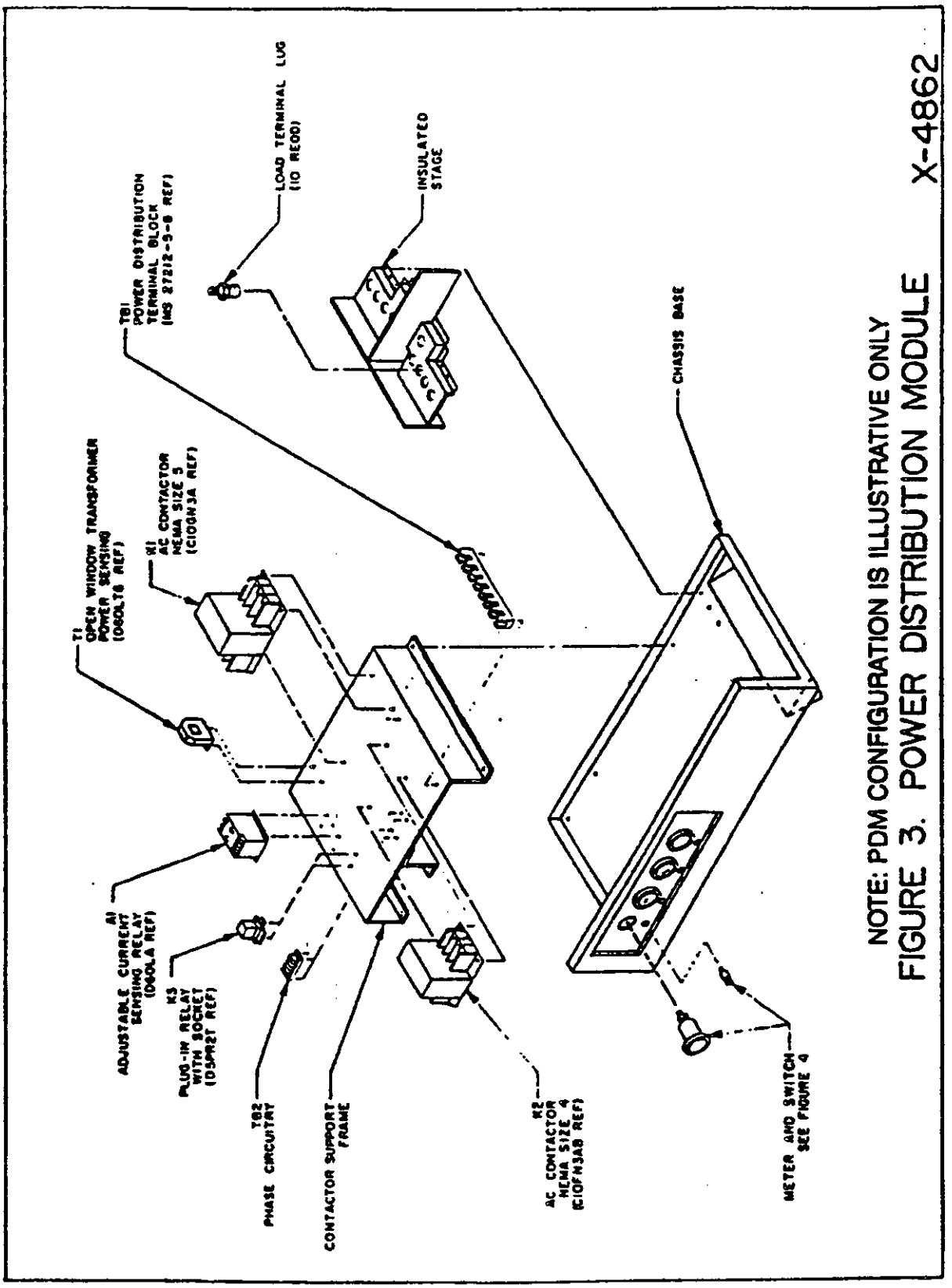
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NOTE: PDM CONFIGURATION IS ILLUSTRATIVE ONLY

FIGURE 2. POWER DISTRIBUTION MODULE X-4861

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X-4862

NOTE: PDM CONFIGURATION IS ILLUSTRATIVE ONLY

FIGURE 3. POWER DISTRIBUTION MODULE

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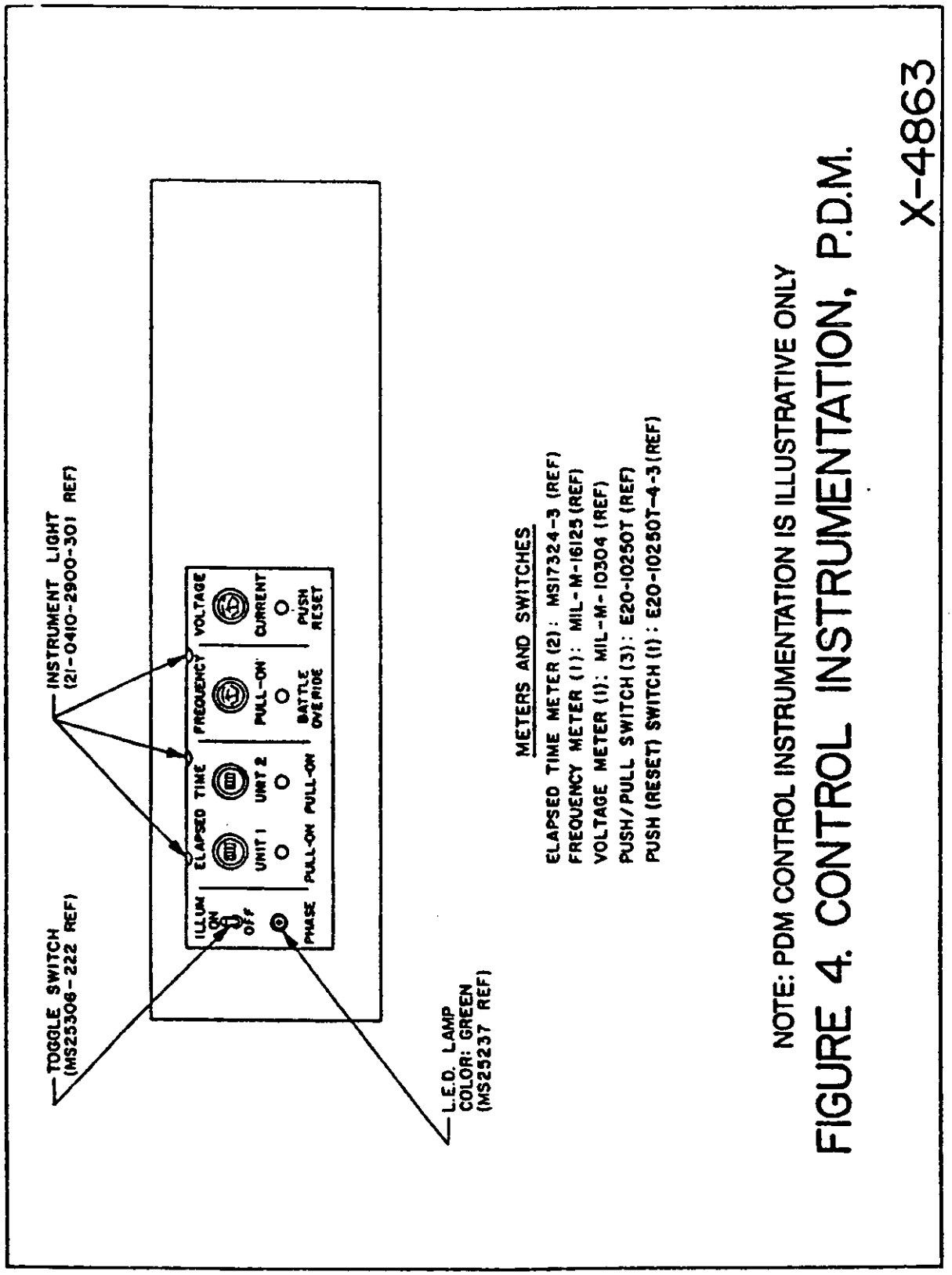


FIGURE 4. CONTROL INSTRUMENTATION, P.D.M.

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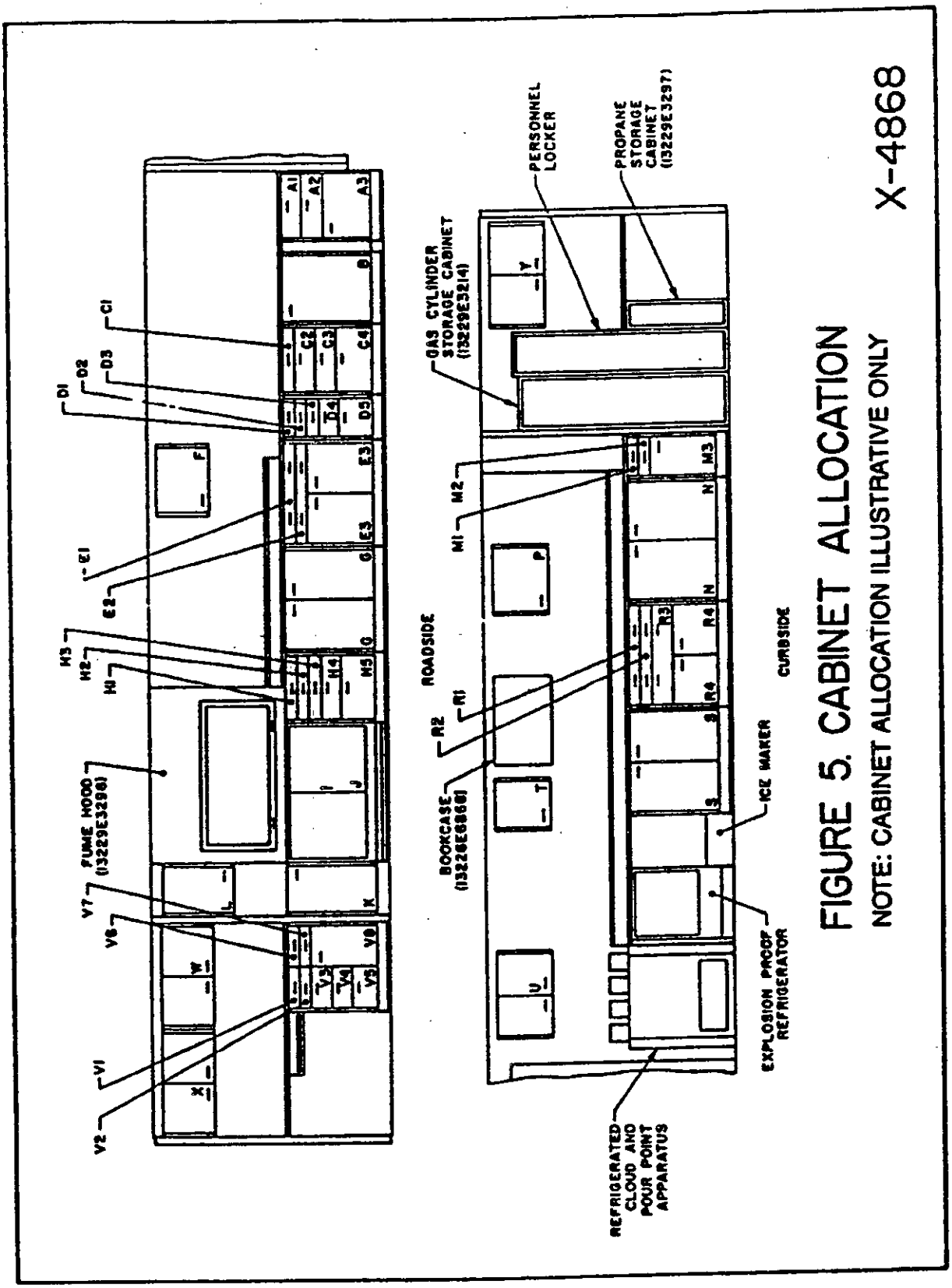
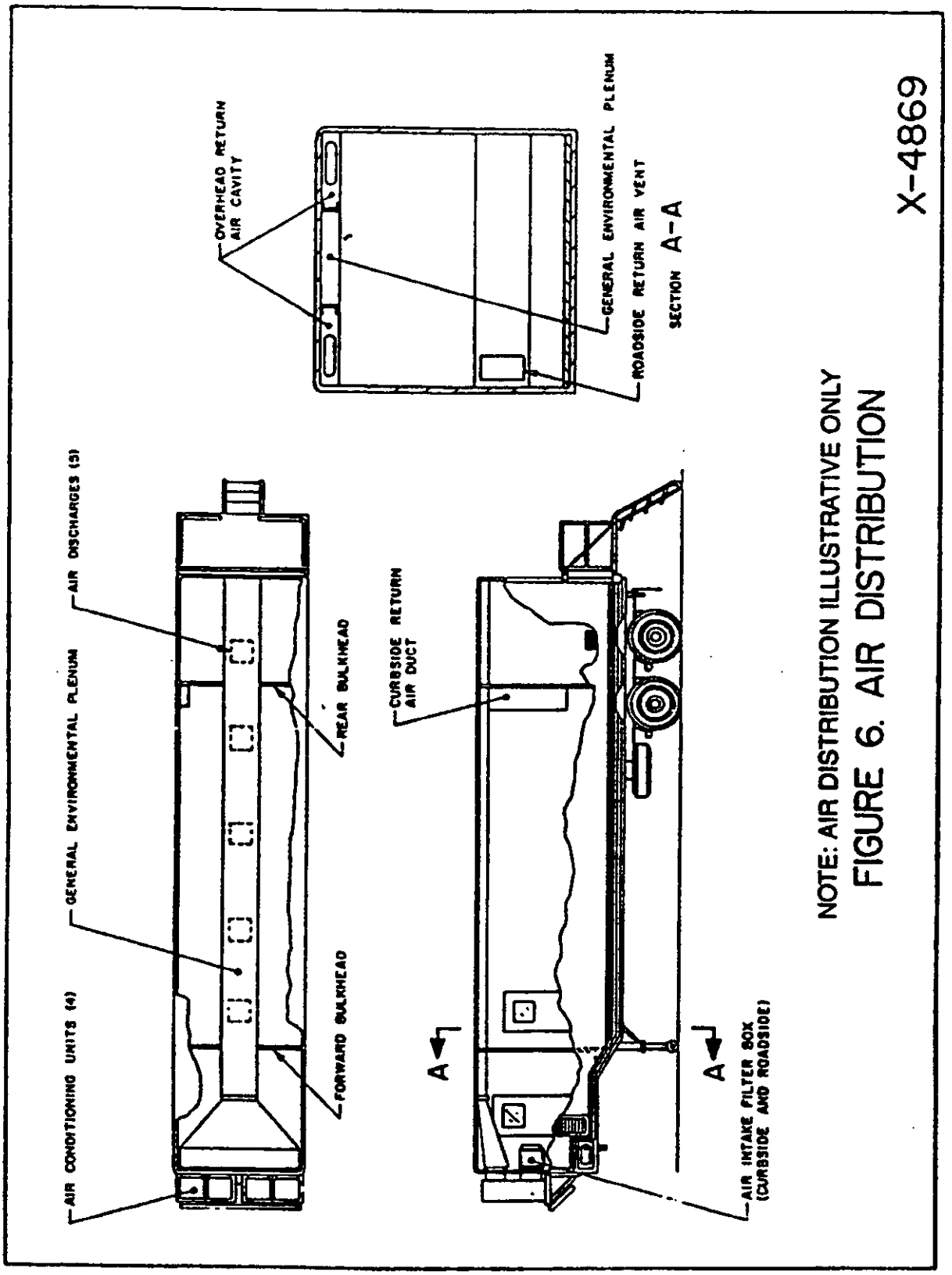


FIGURE 5. CABINET ALLOCATION

NOTE: CABINET ALLOCATION ILLUSTRATIVE ONLY

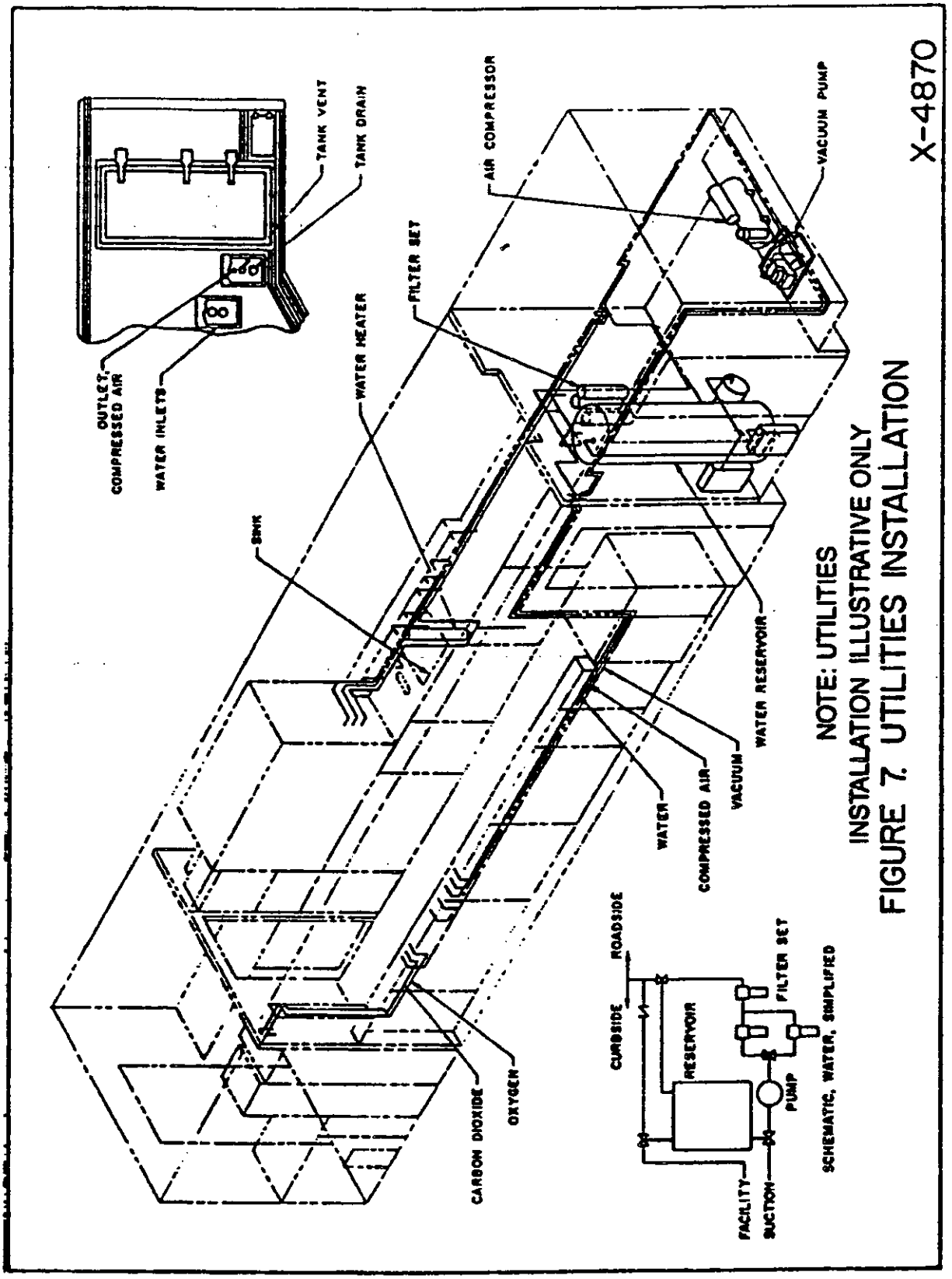
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NOTE: AIR DISTRIBUTION ILLUSTRATIVE ONLY

FIGURE 6. AIR DISTRIBUTION

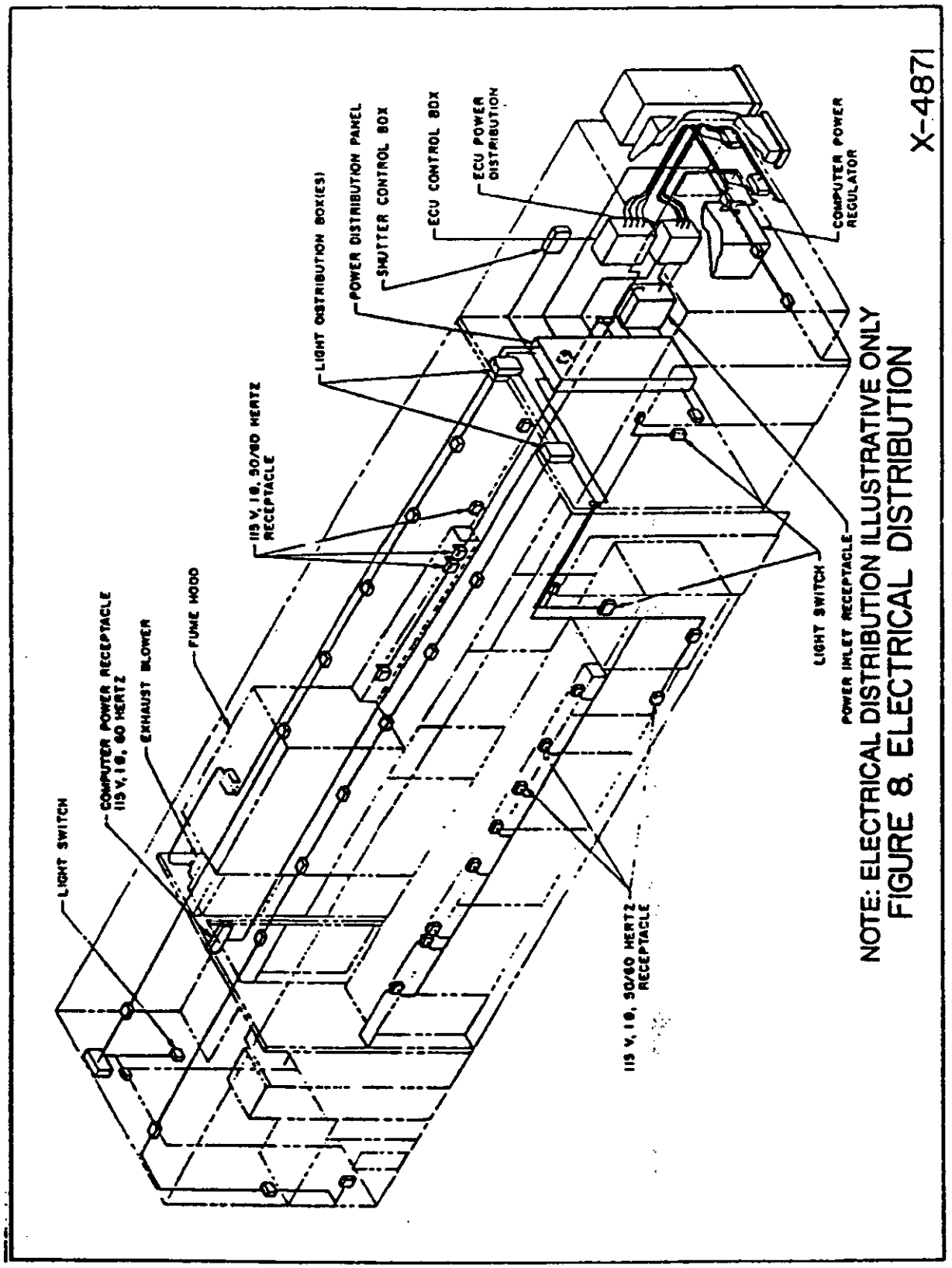
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NOTE: UTILITIES INSTALLATION ILLUSTRATIVE ONLY

FIGURE 7. UTILITIES INSTALLATION

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NOTE: ELECTRICAL DISTRIBUTION ILLUSTRATIVE ONLY

FIGURE 8. ELECTRICAL DISTRIBUTION

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

RECOMMEND A CHANGE	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-L-53110	91/09/12

3. DOCUMENT TITLE

Laboratory, Petroleum, Modular Base

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
	(703) 664-5717	354-5717
c. ADDRESS (Include Zip Code) US Army Belvoir RDE Center ATTN: STRBE-TSE	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466	

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